Indian Institute of Technology, Ropar
Mechanical Engineering
Department

CP302 BTP PROJECT REPORT

Microstructural and Mechanical Characterisation of Cold-Sprayed SS316L Coatings

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Contents

1 Introduction

1.1 Thermal spray

1.2 Types of thermal spray

1.3 Cold spray process and its applications

2. Material and Methods

2.1 SS316l

2.2 Substrate selection

2.3 Cold spray deposition process parameters

2.4 Heat treatment

3. Results and discussions

3.1 Microstructure analysis of the cold spray deposition of the SS316L

3.2 Microhardness and porosity analysis of the Coating

4. Conclusion

Title: Microstructural and Mechanical Characterisation of Cold-Sprayed SS316L Coatings

1 Introduction: -

1.1 Thermal Spray Process: -

Thermal spray is a surface coating process in which a powder or wire feedback material is heated to a molten or semi-molten state and then sprayed onto a substrate at high velocity to form a protective coating.

- The heat source may be a combustion flame, plasma arc, electric arc, or high-velocity oxy-fuel (HVOF) system.
- The molten droplets strike the surface, flatten into thin layers (called splats), cool rapidly, and solidify to build up the coating.
- The substrate itself usually remains relatively cool, so its bulk properties are not significantly affected.
- Coatings produced by thermal spraying can range from a few micrometres to several millimetres in thickness.
- A wide variety of materials can be deposited, including metals, ceramics, polymers, and cermet.
- The process is widely used to improve wear resistance, corrosion resistance, thermal insulation, oxidation resistance and surface hardness.
- It is also used to restore worn or damaged machine components, increasing their service life and reducing replacement costs.

Classification of Thermal Spray Processes: -

1. Flame Spraying
2. Plasma Spraying
3. Arc Spraying
4. HVOF (High Velocity Oxy-Fuel) Spraying.
5. HVAF (High Velocity Air-Fuel)

These processes typically operate at very high temperatures) often above 2000 °C) to melt the feedstock material and propel it toward the substrate at high speed. The resulting coating has strong adhesion and provides excellent protection under severe operating conditions.

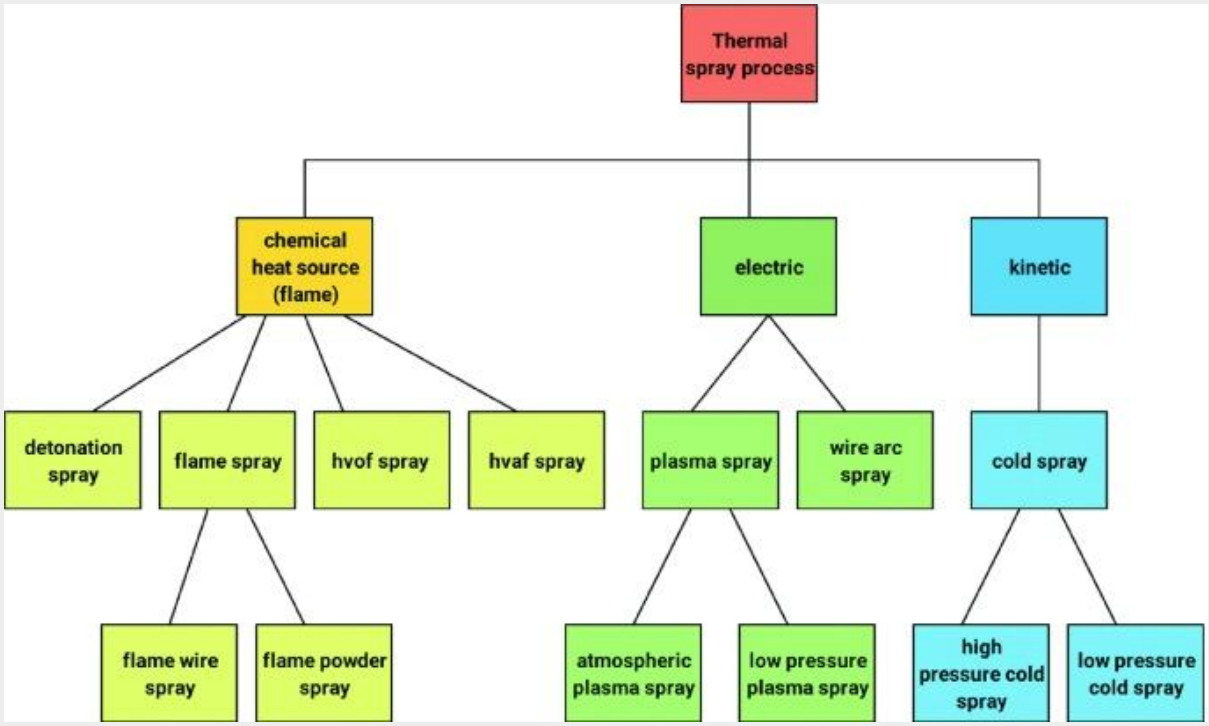


Figure XX. Classification of thermal spray processes based on the energy source, including chemical (flame), electrical, and kinetic spray techniques

Cold Spray: -

Cold spray is an advanced coating technique in which solid powder particles are accelerated to very high (often supersonic) velocities using a high-pressure gas and deposited onto a surface without melting.

- Particles typically travel at 300–1500 m/s through a De Laval nozzle.
- When the particles impact the substrate, they undergo severe plastic deformation and bond due to their kinetic energy.
- Since the particles remain solid throughout the process, oxidation, phase transformation, and thermal damage are minimized.
- Cold spray is therefore classified as a solid-state deposition process.
- The process generally uses gases such as nitrogen, helium, or compressed air as the propellant.
- Materials commonly deposited include aluminum, copper, nickel, titanium, and their alloys.
- The resulting coatings are dense, have low porosity, and retain the original properties of the feedstock material.
- Because no melting occurs, residual stresses and distortion of the substrate are much lower than in conventional thermal spray processes.
- Cold spray is especially suitable for repairing heat-sensitive components and depositing oxygen-sensitive materials.

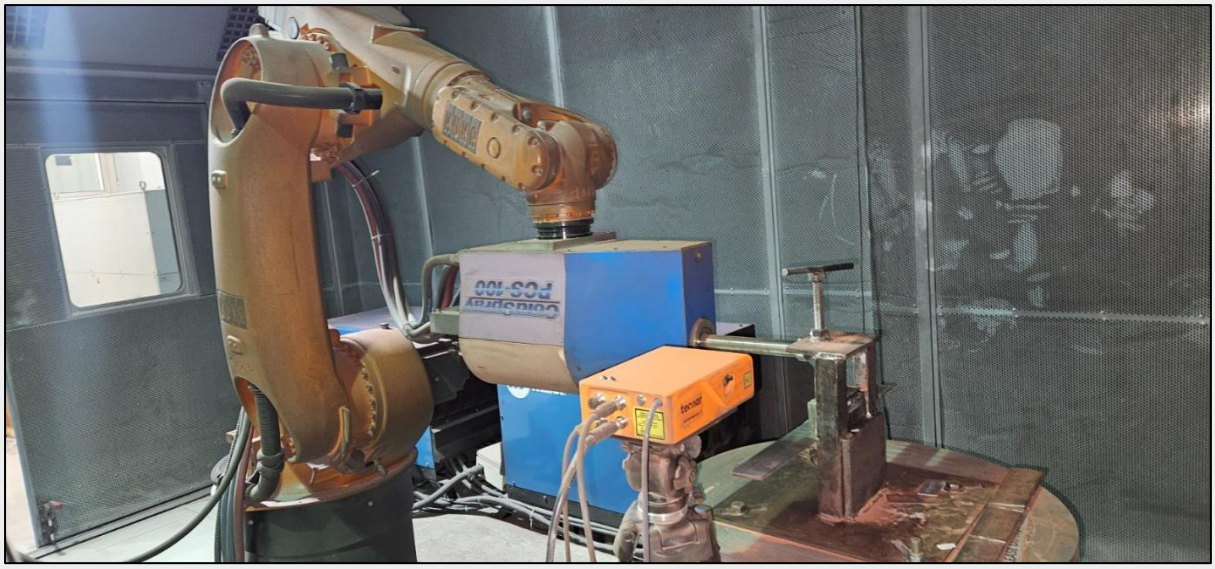


Figure X. Robotic cold spray system employed for coating deposition.

- Types of Cold Spray
    1. High-Pressure Cold Spray (HPCS)
    2. Low-Pressure Cold Spray (LPCS)
- Applications
    - Aerospace component repair and restoration
    - Additive manufacturing
    - Electrical and electronic components
    - Corrosion protection coatings
    - Dimensional restoration of worn parts
    - Biomedical and defense applications

Cold spray is gaining popularity because it can produce high-quality coatings with excellent adhesion while preserving the mechanical and metallurgical properties of both the coating and the substrate.

A. Material and Methods: -

I. SS316L (Stainless Steel 316L) :-

SS316L is a low-carbon version of 316 austenitic stainless-steel containing chromium, nickel, and molybdenum. The letter **"L"** stands for **Low Carbon** (maximum 0.03% carbon), which improves corrosion resistance and reduces the risk of carbide precipitation during welding. SS316L is widely used in industries where excellent corrosion resistance, good mechanical strength, and superior weldability are required.

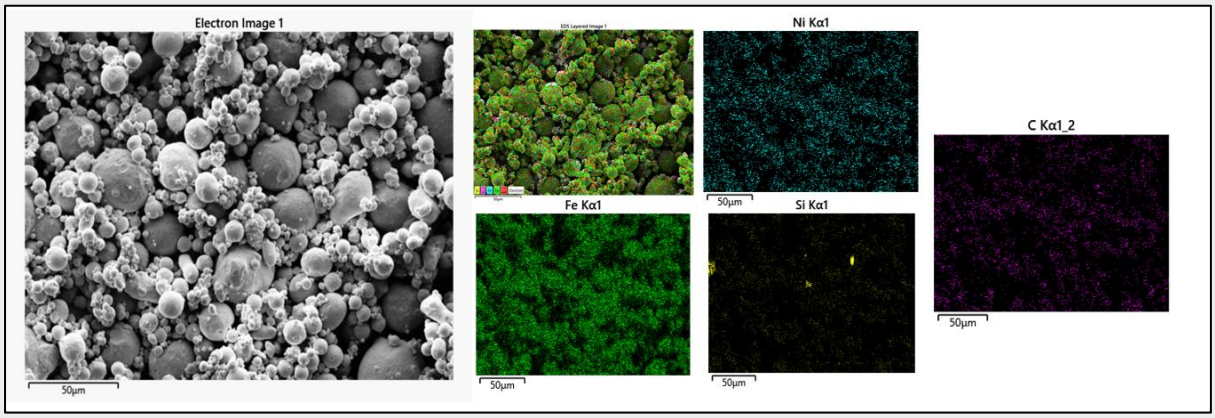


Figure: Figure X. SEM micrograph and EDS mapping analysis of the powder feedstock employed in the cold spray process.

- Key Features of SS316L
  - Excellent resistance to corrosion and oxidation.
  - Superior resistance to pitting and crevice corrosion in chloride environments.
  - Good strength and toughness at both low and high temperatures.
  - Excellent weldability and formability
  - Non-magnetic in the annealed condition.
  - Suitable for use in marine, chemical, and biomedical applications.

- **Chemical Composition of SS316L: -**

Element	Percentage
Chromium (Cr)	16-18%
Nickel (Ni)	10-14%
Molybdenum (Mo)	2-3%
Carbon	0.03%
Manganese	2%
Silicon	0.75%
Iron	Balance

Cr provides corrosion resistance. Ni improves ductility and toughness. Mo increases resistance to pitting corrosion, especially in chloride environment.

II. Feedstock used in the process

- Feedstock Material: Gas-atomized SS316L stainless steel powder was selected as the coating material.
- Powder Particle Size: The powder particle size range was 15–65 µm.
- Substrate Material: A stainless steel substrate was chosen for coating deposition.
- Deposition Technique: The coating was deposited using the Cold Spray (CS) process.
- Coating Preparation: The SS316L powder was supplied as feedstock and sprayed onto the substrate to produce the coating.
- Heat Treatment: After deposition, the coated samples were heat treated at: 750 and 1100 C

- Hardness Testing: Hardness tests were carried out on the as-sprayed and heat-treated coatings to evaluate the effect of heat treatment on coating hardness.

III. Cold Spray Deposition Parameters: -

The properties of the cold-sprayed SS316L coating are influenced by several process parameters:

Parameters	Value
Pressure (P)	4.5 MPa
Temperature (T)	750C
Stand of Distance (SOD)	30 Cm
Transverse Speed	100mm/s

These parameters significantly affect coating characteristics such as adhesion strength, porosity, hardness, thickness, and overall coating performance.

Heat Treatment: -

Heat treatment is a process in which the coated samples are heated to a specific temperature and held for a certain period of time to modify their microstructure and properties. In this study, the cold-sprayed SS316L coated samples were heat treated at: 750°C for 1 hour and 1100°C for 1 hour. After holding at the required temperature, the samples were allowed to cool to room temperature.

Purpose of Heat Treatment

- Relieve residual stresses developed during cold spray deposition.
- Improve bonding between deposited particles.
- Reduce porosity and defects in the coating.
- Promote diffusion between coating particles and substrate.
- Improve ductility and overall mechanical properties.

1. Purpose of the Etching: -

Etching is a metallographic technique used to reveal the microstructure of a polished sample by applying a chemical reagent. After polishing, the coated samples were etched using a suitable stainless-steel etchant (aqua regia)

Purpose of Etching of the coating

- Reveal intersplat boundaries.
- Show particle boundaries and coating microstructure.
- Identify pores and defects.
- Observe changes after heat treatment.

B. Result and Discussion:-

Microstructure

1) Scanning Electron Microscopy (SEM)

Scanning Electron Microscopy (SEM) is a characterization technique used to observe the surface morphology and microstructure of materials at very high magnification. In SEM, a focused beam of electrons scans the sample surface and produces detailed images. The as-sprayed cold-sprayed coating exhibited severely deformed particles due to the high-velocity impact of powder particles during deposition. Distinct inter-splat boundaries and localized porosity were observed in the coating cross-section. After heat treatment, the inter-splat porosity was noticeably reduced because of enhanced diffusion and improved particle bonding. The coating heat treated at 750 °C for 1 h showed partial reduction in porosity; however, some inter-particle gaps were still visible. In contrast, the sample heat treated at 1100 °C for 1 h exhibited a denser microstructure with significantly reduced porosity and less prominent splat boundaries. The higher heat treatment temperature promoted better metallurgical bonding and diffusion across particle interfaces. Nevertheless, a few residual pores were still present, indicating that longer heat treatment duration and/or higher thermal exposure may be required for complete pore elimination and full densification of the coating.

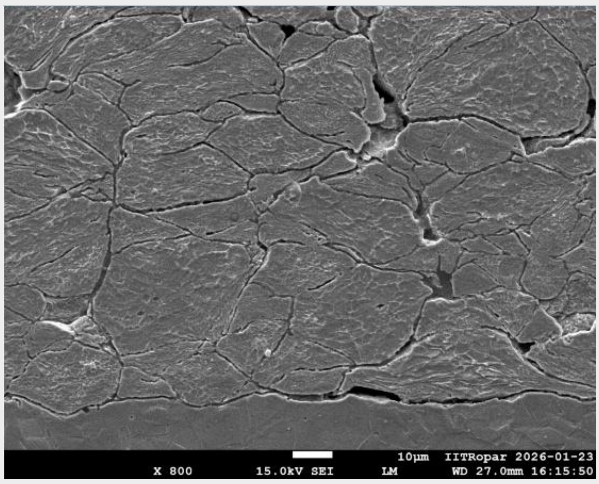


Figure X. Etched cross-sectional microstructure of the as-sprayed cold-sprayed coating showing inter-splat boundaries and particle deformation.

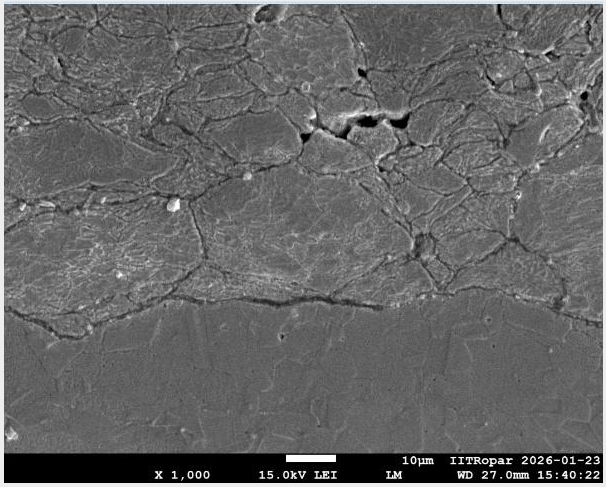


Figure X. Etched cross-sectional microstructure of the cold-sprayed coating heat treated at 750 °C for 1 h, showing inter-splat boundaries and microstructural evolution.

- 1100°C heat-treated coating: Lowest porosity and most dense microstructure.

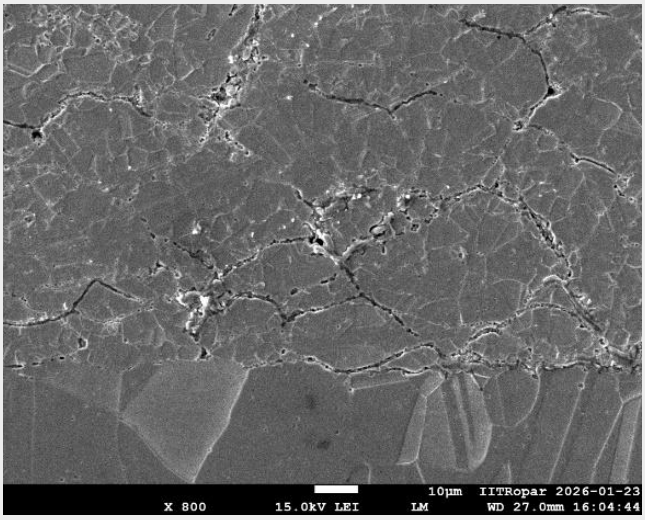


Figure X. Etched cross-sectional microstructure of the cold-sprayed coating heat treated at 1100 °C for 1 h, showing diffusion across inter-splat boundaries and microstructural changes.

2) Porosity: -

The as-sprayed SS316L coating exhibited the highest porosity because the deposited particles were bonded mainly through mechanical and less deformed particles, leaving visible gaps between particles. After heat treatment at 750°C for 1-hour, partial diffusion between particles improved bonding and reduced some of the pores, resulting in a more compact microstructure. After heat treatment at 1100°C for 1-hour, extensive diffusion and strong metallurgical bonding significantly reduced porosity, producing the densest and most uniform coating among all three conditions.

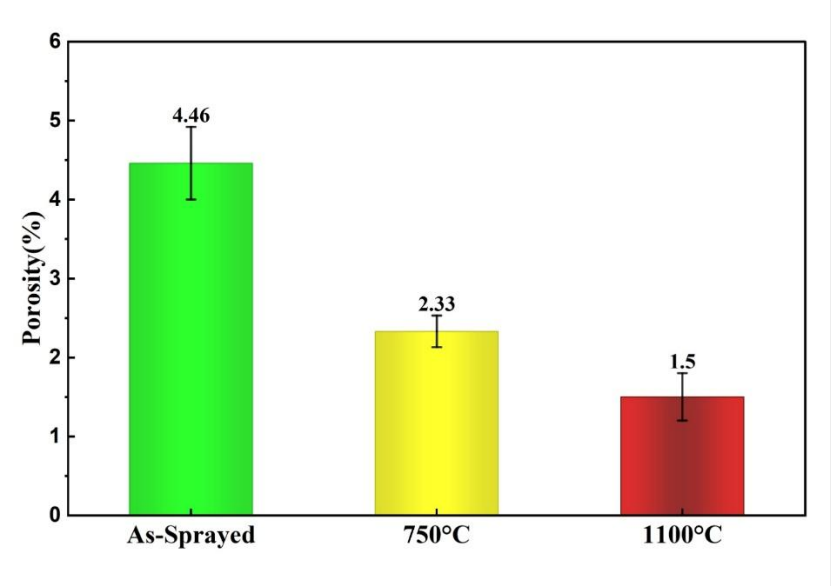


Figure X. porosity variation of the cold-sprayed coating in the as-sprayed and heat-treated conditions (750 °C for 1 h and 1100 °C for 1 h).

Mechanical Characterisation

3) Mechanical Characteristics (Microhardness) :-

Microhardness is the resistance of a material to indentation or localized deformation, usually measured using a Vickers microhardness tester. In the as-sprayed cold-sprayed SS316L coating, microhardness is relatively high because the powder particles undergo severe plastic deformation during impact. After heat treatment at 750°C for 1 hour, particle bonding improves and porosity decreases, so the hardness may remain the same or increase slightly. After heat treatment at 1100°C for 1 hour, the coating becomes denser, but recrystallization and grain growth soften the material, which generally results in lower microhardness.

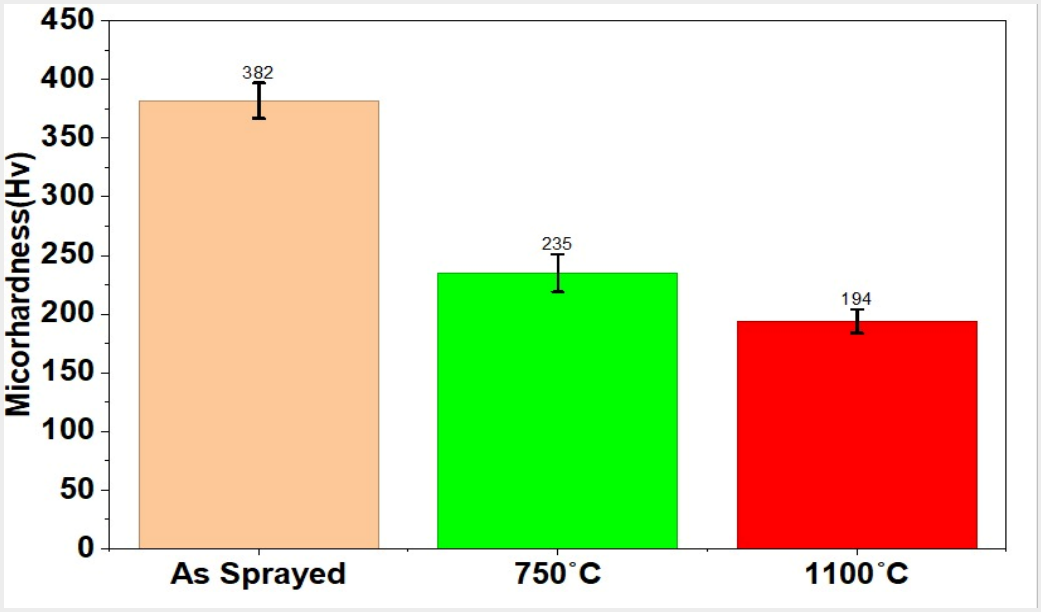


Figure X. Microhardness variation of the cold-sprayed coating in the as-sprayed and heat-treated conditions (750 °C for 1 h and 1100 °C for 1 h).

Conclusions:

• Successful deposition of SS316L coating on the SS316L substrate was achieved using the cold spray process, producing a dense and well-adhered coating with significant particle deformation.
• The as-sprayed coating exhibited distinct inter-splat boundaries and localized porosity due to incomplete bonding between deposited particles during high-velocity impact.
• Heat treatment at 750 °C and 1100 °C for 1 h improved the inter-particle bonding and reduced the inter-splat porosity through enhanced diffusion.
• The as-sprayed coating showed higher hardness because of severe plastic deformation and strain hardening, whereas heat treatment resulted in a gradual reduction in hardness due to stress relaxation, recrystallization, and grain growth.
• Overall, post-heat treatment significantly enhanced the coating densification and metallurgical bonding, indicating that thermal treatment is an effective approach for improving the coating and performance of cold-sprayed SS316L coatings.

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